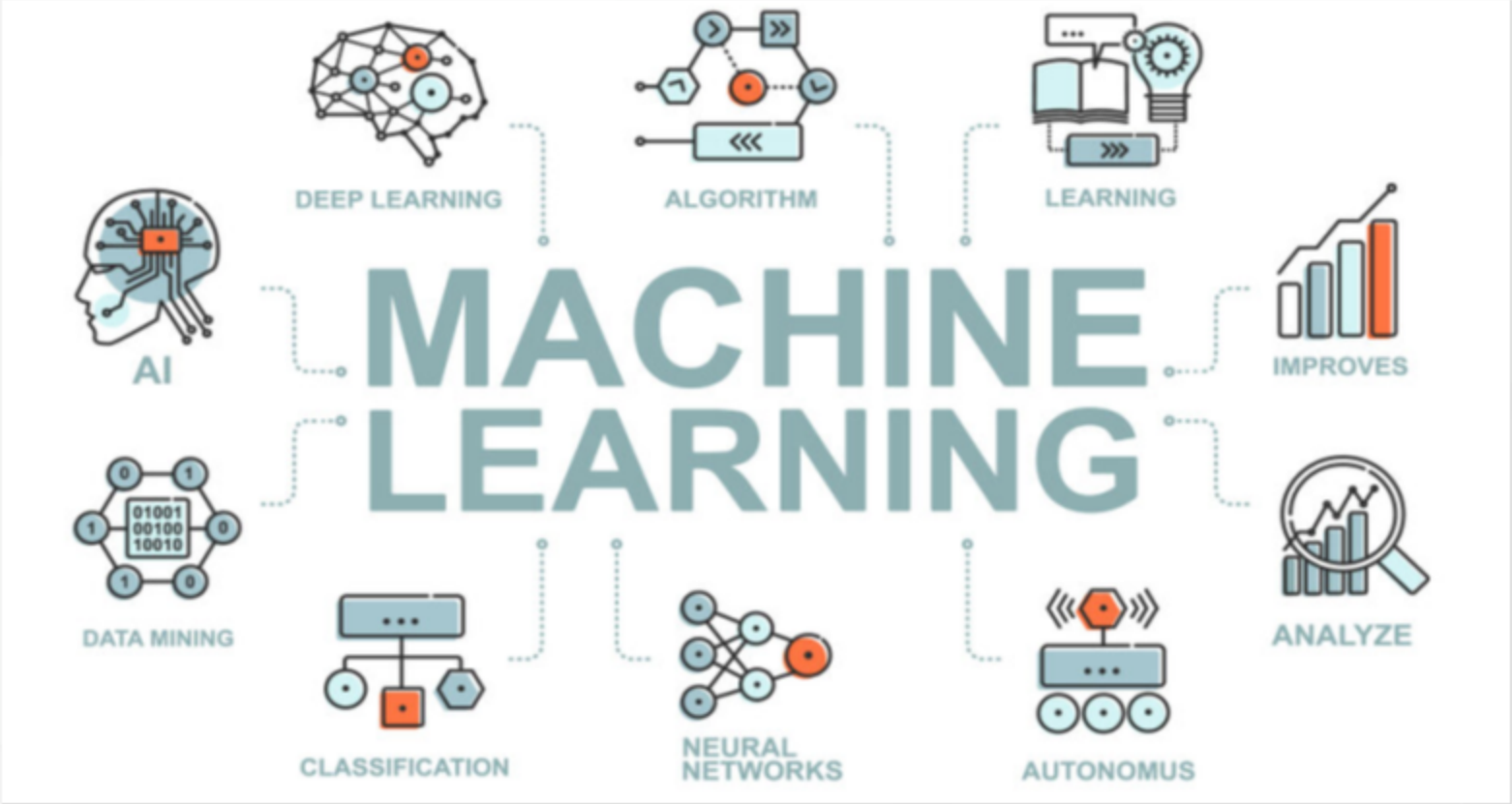
Introduction to Machine Learning

Agenda

- What is Machine Learning?
- When do we use Machine Learning?
- Key Concepts in Machine Learning
- Types of Machine Learning
- Applications of Machine Learning
- Challanges in Machine Learning

What is Machine Learning?

Machine learning is a subset of artificial intelligence (AI) that focuses on the development of algorithms and models that enable computers to learn and make predictions or decisions without being explicitly programmed to perform a specific task.



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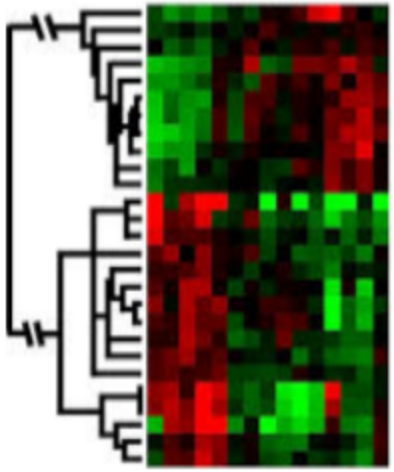
When do we use Machine Learning?

Machine Learning is used when:

- Human expertise does not exist (navigating on Mars)
- Humans can't explain their expertise (speech recognition)
- Models must be customized (personalized medicine)
- Models are based on huge amounts of data (genomics)



Key Concepts in Machine Learning

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- **Data:**

Data is the foundation of Machine Learning. It can be any type of information, such as text, numbers, images, or sensor readings, collected from various sources.

- **Algorithm:**

An algorithm is a set of instructions that the ML system follows to learn from data.

- **Training:**

Training is the process of teaching a machine learning model by providing it with data.

- **Testing & Evaluation:**

After training, the model is tested on new, unseen data to assess its performance.

- **Prediction & Inference:**

Once trained, the ML model can make predictions or classifications on new, unseen data.

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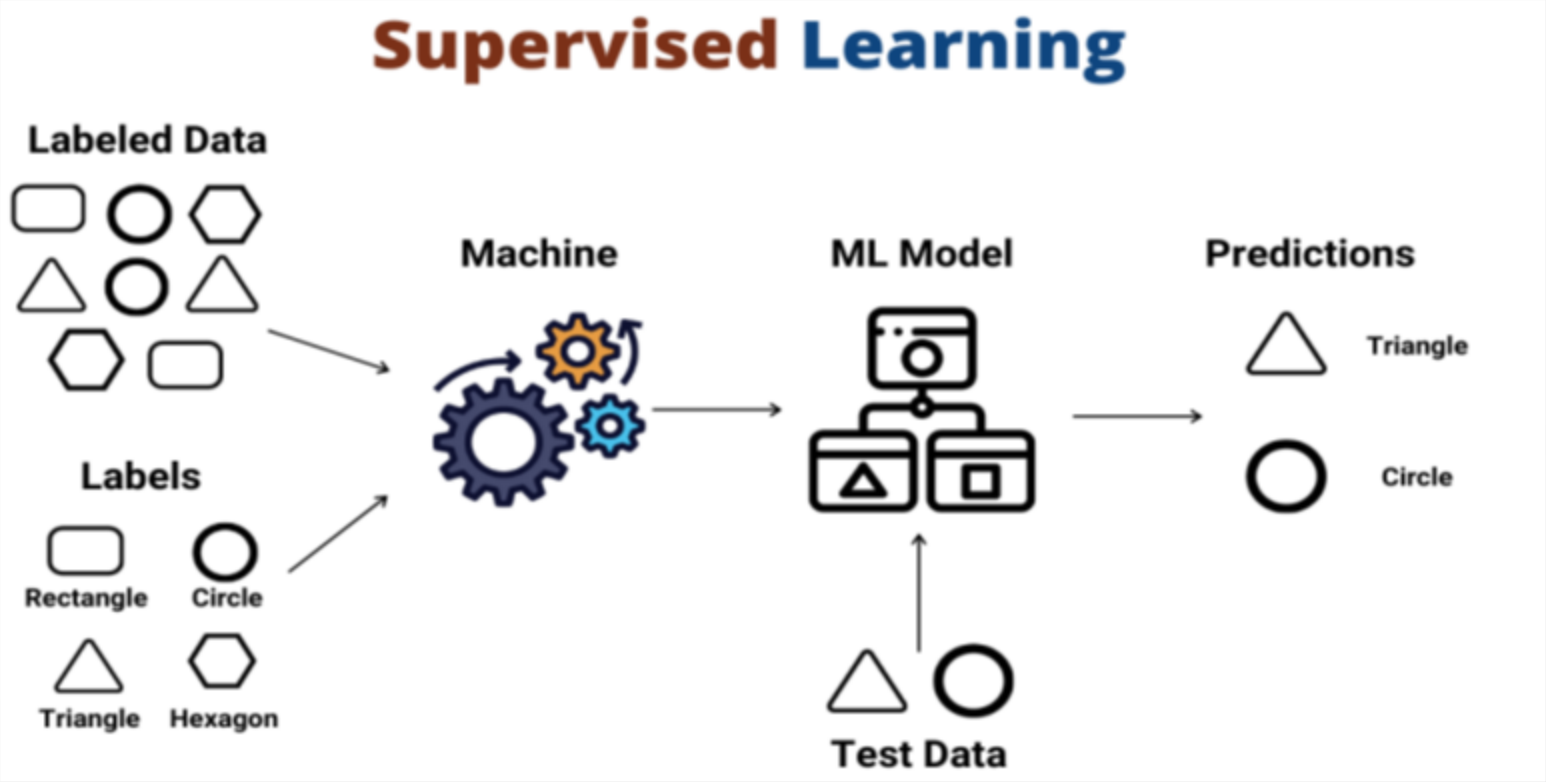
Types of Machine Learning

Machine Learning can be categorized into three main types:

1. Supervised Learning:

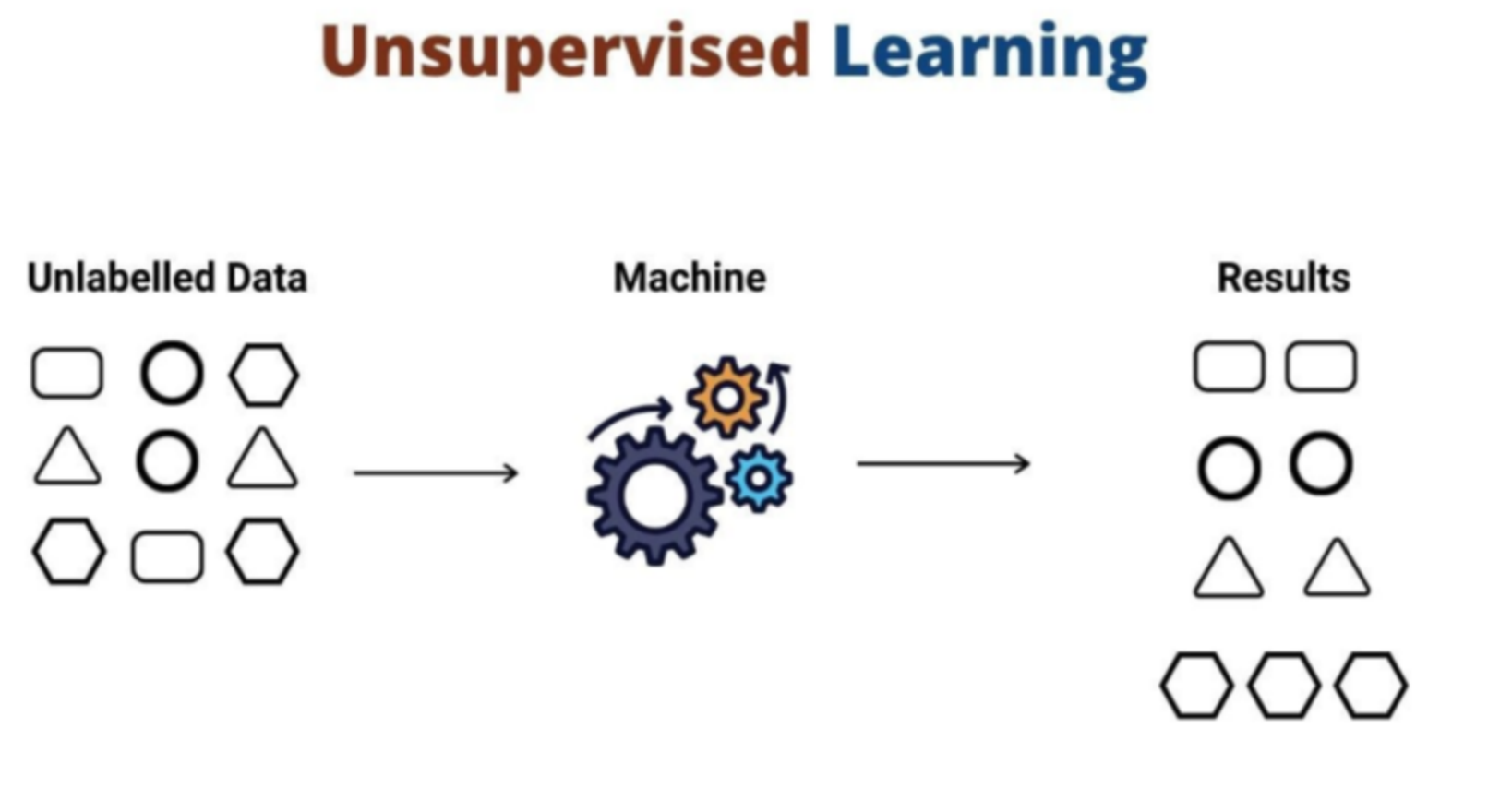
In supervised learning, the model is trained on labeled data, meaning that each input is associated with a known output.



Types of Machine Learning

2. Unsupervised Learning:

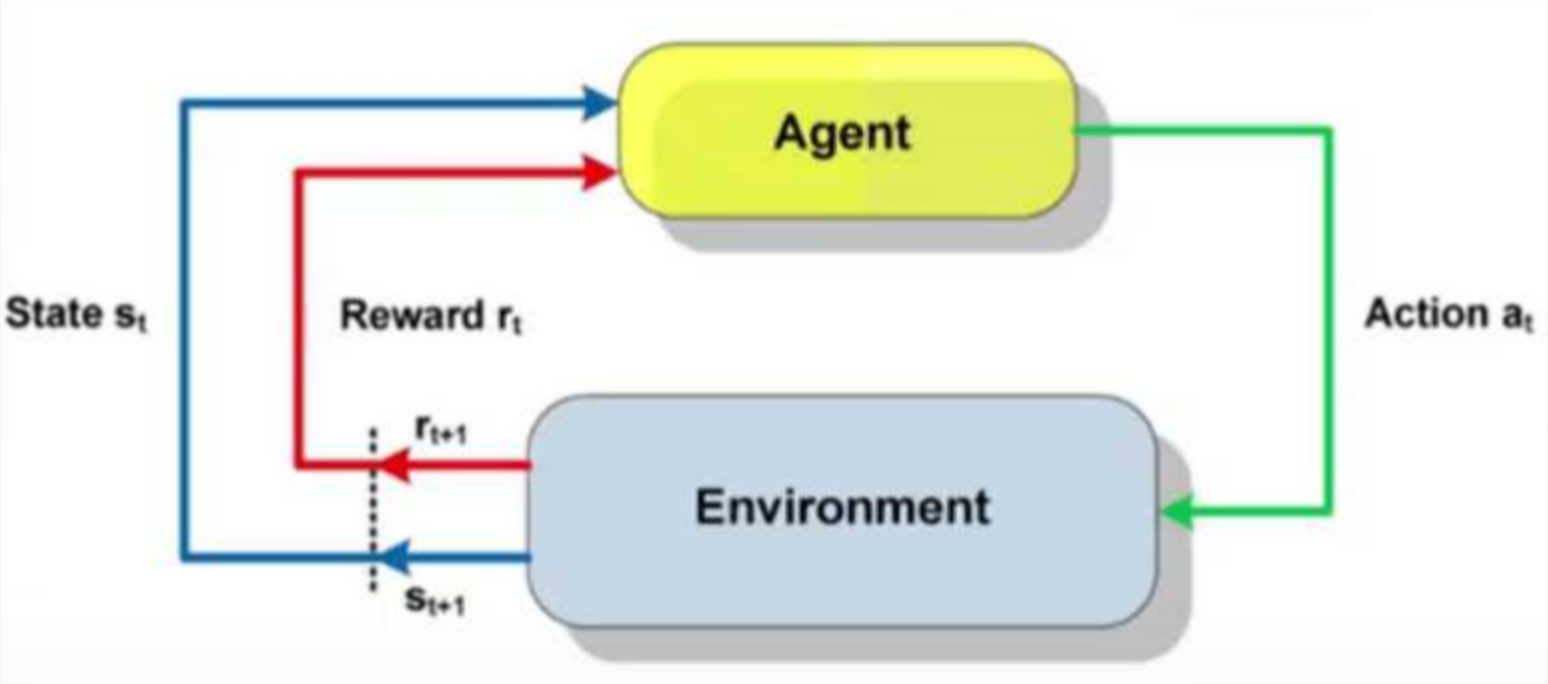
Unsupervised learning deals with unlabeled data. The model's goal is to discover patterns, structures, or groupings within the data.



Types of Machine Learning

3. Reinforcement Learning:

Reinforcement learning involves training a model to make a sequence of decisions in an environment to maximize a reward. This is used in applications like autonomous driving and game-playing AI.



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Applications of Machine Learning

- Healthcare
- Finance
- Recommendation System
- Autonomous Vehicle
- Manufacturing

Challanges in Machine Learning

- Data Quality
- Overfitting & Underfitting
- Interpretability
- Ethical Concerns